

KRISHNA SAI VARDHAN KANTIPUDI

Data Engineer | AI/ML

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SUMMARY

Data Engineer (AI/ML) with 3+ years of experience designing and optimizing scalable **ETL/ELT pipelines**, real-time data processing, and AI-powered analytics solutions across enterprise networking and banking domains. Proficient in **Python, Apache Spark, Kafka, Databricks, Snowflake, Azure, AWS, and SQL**, building cloud-native data platforms that support machine learning, network telemetry analytics, and business intelligence. Experienced in developing distributed data pipelines, feature engineering workflows, and MLOps solutions using **MLflow, Airflow, Docker, and Kubernetes** while collaborating with cross-functional engineering teams in Agile environments. Passionate about leveraging **AI, Big Data, and cloud technologies** to build reliable, high-performance enterprise data solutions.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, C, C++

Big Data & Distributed Processing: Apache Spark (PySpark, MLlib), Apache Kafka, Databricks, Apache Airflow, dbt

Cloud Platforms: Microsoft Azure, AWS

Databases & Data Warehousing: Snowflake, MySQL, Vector Databases, Database Query Optimization, Data Migration

AI/ML Frameworks & Libraries: TensorFlow, PyTorch, Scikit-learn, MLflow, LangChain, Azure OpenAI, LLM Integration, OpenCV (CNN)

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Plotly, EDA, Predictive Modeling

DevOps & CI/CD: Docker, Kubernetes, Git, CI/CD Pipelines

Tools & Platforms: Jupyter Notebook, Streamlit, NVIDIA GPU Platforms, Windows

Concepts & Methodologies: Machine Learning (Supervised/Unsupervised), Data Cleaning & Preprocessing, ETL/ELT, Feature Engineering, Agile Scrum

Core Strengths: Cross-Functional Collaboration, Stakeholder Communication, Analytical Problem-Solving, Technical Leadership, Continuous Learning

PROFESSIONAL EXPERIENCE

Data Engineer (AI/ML) | Cisco Systems, San Jose, CA, USA

Jul 2025 – Present

- Engineered scalable ETL and streaming data pipelines using **Apache Spark, Kafka, and Databricks** to process high-volume **network telemetry, device logs, and security events**, reducing end-to-end data latency by **40%** and enabling real-time AI-driven network analytics.
- Built feature engineering pipelines with **PySpark** and **Snowflake** to transform large-scale networking and infrastructure datasets into machine learning-ready features, improving predictive model performance and reducing feature preparation time by **30%**.
- Developed automated MLOps workflows using **Docker, Kubernetes, Airflow, and MLflow** to streamline model training, validation, and deployment for AI-powered network monitoring solutions, reducing release cycles by **45%**.
- Supported enterprise **AI Assistant** initiatives by integrating **LLM-based semantic search** and **retrieval-augmented generation (RAG)** over internal networking knowledge repositories, improving troubleshooting accuracy and accelerating engineer response times by **25%**.
- Collaborated with **Network Engineering, Security, Platform Engineering, and AI/ML** teams in Agile environments to build scalable cloud-native data platforms on **Azure** and **AWS**, supporting production AI applications across enterprise networking and security operations.
- Optimized large-scale **Snowflake** and **Databricks** workloads through partition optimization, clustering strategies, and query tuning, lowering cloud compute costs by **20%** while processing terabytes of daily network telemetry.
- Built monitoring datasets and analytics pipelines supporting **AI-driven anomaly detection** for enterprise network infrastructure, improving detection precision by **18%** and enabling proactive identification of network performance and security issues.

Data Engineer | Mphasis, India

Jan 2022 – Jul 2024

- Developed fraud detection models and pipelines using **PySpark, Databricks, and Azure OpenAI**, processing millions of banking transactions daily and reducing false positives by **22%**.
- Built customer analytics and recommendation systems leveraging **TensorFlow, PyTorch, and LangChain**, driving a **15%** increase in personalized product engagement for retail banking clients.
- Designed and implemented a centralized Data Lake architecture on **Azure** and **AWS** using **dbt** and **Airflow**, consolidating **10+** disparate banking data sources and improving data accessibility by **35%**.
- Migrated legacy ETL workflows to **Databricks** and **Snowflake** using **PySpark** and **SQL**, reducing batch processing time by **40%** and eliminating manual data reconciliation errors.
- Implemented **MLflow**-based model tracking and versioning for GenAI and banking AI initiatives, improving model reproducibility and reducing deployment errors by **30%**.
- Designed and optimized **Apache Kafka** streaming pipelines for **real-time transaction monitoring**, integrating automated **data quality** and **data governance** validations using **Python** and **SQL** to enable **sub-second fraud detection**, improve **data accuracy by 25%**, and ensure **banking regulatory compliance**.

PROJECTS

Super Store Data Analytics Platform | Teachnook (via Webin)

- Developed an end-to-end **Python-based ETL and analytics pipeline** to process retail sales data, performing data cleaning, feature engineering, and exploratory analysis using **Pandas** and **NumPy**.
- Built an interactive **Streamlit** dashboard to visualize business KPIs and profitability trends, enabling data-driven insights for sales, inventory, and customer performance.

AI-Powered Music Recommendation System | Capstone Project

- Developed a machine learning recommendation system using **TensorFlow, OpenCV, and Python** to classify facial emotions and deliver personalized music recommendations in real time.
- Built scalable data preprocessing and model inference pipelines, improving prediction accuracy and demonstrating end-to-end AI application development.

Big Data & Machine Learning Applications | University of North Texas

- Built distributed machine learning pipelines using **Apache Spark MLlib**, implementing classification and clustering algorithms on large-scale datasets.
- Applied feature engineering, model evaluation, and hyperparameter tuning to optimize predictive performance using **Spark, Python, and SQL**.

EDUCATION

Master of Computer Science and Information Sciences

University of North Texas., Denton, TX | USA

Aug 2024 – May 2026

Bachelor of Technology in Computer Science Engineering (Artificial Intelligence)

Parul University, Vadodara | India

Nov 2020 – Apr 2024